

OTTAWA CDA, ON, Canada

WMO: 710630

Lat: 45.3825N

Lon: 75.7139W

Elev: 260

StdP: 14.56

Time Zone: -5.00 (NAE)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.5	-5.5	-19.4	1.9	-9.5	-13.7	2.6	-4.2	20.9	20.6	19.0	16.7	6.4	270	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	87.7	71.3	84.4	69.8	81.6	68.7	74.8	82.9	72.8	80.4	71.1	78.4	9.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.3	120.9	78.8	70.4	113.0	77.2	68.6	106.3	75.4	38.5	82.9	36.7	80.3	35.2	78.5	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.8	16.7	15.0	-15.6	92.8	5.2	2.9	-19.3	94.9	-22.3	96.7	-25.2	98.3	-28.9	100.5		
			-16.1	77.9	5.0	2.6	-19.7	79.7	-22.6	81.3	-25.4	82.7	-29.1	84.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.5	15.2	17.8	28.0	43.2	56.7	65.5	70.3	68.5	61.0	47.9	35.8	22.8
	DBStd	21.25	12.31	10.53	10.87	8.91	7.85	6.52	5.25	5.41	7.23	8.10	8.72	10.83
	HDD50	4342	1077	900	685	240	25	0	0	0	3	140	430	842
	HDD65	7928	1542	1320	1146	655	278	73	12	26	162	531	876	1307
	CDD50	2350	0	0	3	36	232	464	629	574	334	75	3	0
	CDD65	463	0	0	0	2	20	87	175	135	43	1	0	0
	CDH74	3680	0	0	2	24	213	702	1401	975	349	14	0	0
	CDH80	961	0	0	0	3	48	184	401	246	78	1	0	0
(14) Wind	WSAvg	7.2	7.5	7.9	8.2	8.6	7.6	6.5	6.0	5.9	5.9	6.9	7.4	7.6
(15) Precipitation	PrecAvg	37.0	2.8	2.1	2.4	3.0	3.0	3.7	3.5	3.3	3.6	3.4	2.9	3.1
	PrecMax	46.5	4.9	4.1	5.0	6.1	5.0	7.6	7.7	6.4	6.3	7.2	4.4	5.4
	PrecMin	31.2	1.2	0.9	0.9	0.6	1.1	0.7	0.7	1.5	1.7	0.9	0.9	1.0
	PrecStd	3.7	1.0	0.8	1.1	1.3	1.1	1.5	1.7	0.9	1.3	1.3	0.9	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	43.0	63.5	77.3	85.4	89.5	91.8	91.2	87.3	75.8	62.0	50.4
		MCWB	44.4	40.0	55.5	61.5	69.4	73.0	74.0	72.3	72.3	64.5	55.1	46.9
	2%	DB	39.9	39.2	51.8	69.2	79.7	85.2	87.8	85.7	81.6	69.9	57.0	44.7
		MCWB	37.9	35.5	44.8	55.3	64.3	70.3	71.4	70.8	69.6	61.5	52.5	42.5
	5%	DB	36.3	36.5	45.6	63.1	75.2	81.2	84.3	82.3	77.9	65.2	52.0	39.8
		MCWB	34.6	34.0	39.9	52.0	61.1	68.2	69.9	68.9	67.2	58.1	47.4	37.8
	10%	DB	32.9	33.8	41.8	57.7	71.1	77.9	81.4	79.5	74.3	60.6	48.2	36.5
		MCWB	31.4	31.9	37.2	48.5	58.9	66.2	69.0	67.9	65.6	54.9	44.0	34.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.6	40.8	56.1	63.5	71.8	76.4	78.3	76.5	74.1	68.0	57.4	47.4
		MCDB	44.9	42.9	62.9	73.3	81.4	83.9	87.2	83.8	82.9	72.9	60.1	48.9
	2%	WB	38.4	36.4	45.9	58.2	67.8	73.5	75.2	73.3	71.4	62.6	52.5	42.8
		MCDB	39.5	38.3	50.9	66.1	76.2	81.1	83.3	80.7	79.2	68.6	56.2	45.0
	5%	WB	34.6	34.2	40.8	53.4	64.4	70.8	73.1	71.7	69.1	58.9	48.1	38.0
		MCDB	36.0	35.9	44.9	61.0	71.3	78.8	80.4	78.9	76.1	64.1	51.5	39.5
	10%	WB	31.6	31.9	37.7	49.5	61.0	68.3	71.1	70.0	66.6	55.8	44.6	34.9
		MCDB	33.0	33.8	41.1	56.9	68.5	74.6	78.1	76.3	72.4	59.6	47.5	36.2
(35) Mean Daily Temperature Range	5% DB	MDBR	14.9	16.1	15.8	17.5	18.9	17.9	18.3	18.2	19.0	15.2	13.2	12.3
		MCDBR	16.0	16.8	19.7	27.9	25.4	22.7	22.0	22.1	23.2	22.6	19.2	14.2
		MCWBR	15.5	14.9	13.7	16.4	13.4	11.2	10.0	10.2	12.5	14.6	15.2	12.8
	5% WB	MCDBR	15.4	15.8	18.1	24.4	21.2	19.5	18.5	18.5	21.0	20.9	18.3	14.0
		MCWBR	15.2	14.7	14.2	16.5	13.6	11.7	10.6	10.0	12.5	15.1	15.7	13.0
(40) Clear-Sky Solar Irradiance	taub		0.264	0.288	0.325	0.346	0.382	0.415	0.419	0.401	0.371	0.340	0.310	0.279
	taud		2.379	2.302	2.350	2.412	2.349	2.270	2.265	2.321	2.397	2.480	2.492	2.393
	Ebn at Noon		272	284	287	290	281	271	268	269	268	263	252	252
	Edh at Noon		23	30	34	35	38	42	41	38	32	26	21	21
(44) All-Sky Solar Radiation	RadAvg		441	748	1166	1423	1648	1793	1821	1605	1256	741	452	323
	RadStd		52	72	100	143	160	139	150	77	105	66	52	28

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page